Linear Algebra and Basics

→ $e^M = \begin{bmatrix} e^{M_{00}} & e^{M_{01}} \\ e^{M_{10}} & e^{M_{11}} \end{bmatrix}$

$M = (M_{ij})_{2\times 2}$

↳ $\begin{pmatrix} M_{00} & M_{01} \\ M_{10} & M_{11} \end{pmatrix}$

$e^M = \left(e^{M_{ij}}\right) \qquad M = (M_{ij})$

→ $e^{i\theta(\vec{n}\cdot\vec{\sigma})} = \cos\theta \cdot I + i\sin\theta\,(\vec{n}\cdot\vec{\sigma})$

$\vec{n}\cdot\vec{\sigma} = \sum_{i=1}^{3} n_i \sigma_i$

→ $\sum_{i=1}^{3} |n_i|^2 = 1$

↳ For pure state

$\rightarrow f: \mathbb{C} \rightarrow \mathbb{C}$ $\quad |\vec{n}| = 1$ $\quad \theta \in \mathbb{R}$ $\quad \vec{n} \rightarrow$ 3-D

$$f(\theta \cdot \vec{n} \cdot \vec{\sigma}) = \left(\frac{f(\theta) + f(-\theta)}{2} \right) \cdot I + \left(\frac{f(\theta) + f(-\theta)}{2} \right) \cdot \vec{n} \cdot \vec{\sigma}$$

$$\rightarrow \varepsilon_{ijk} = \begin{cases} +1 & \text{if } (i,j,k) \text{ is an even permutation of } (1,2,3) \\ -1 & \text{if } (i,j,k) \text{ is an odd permutation of } (1,2,3) \\ 0 & \text{if any 2 indices are equal} \end{cases}$$

$$\sigma_i \sigma_j = \delta_{ij} I + i \sum_k \varepsilon_{ijk} \sigma_k$$

$$\rightarrow \mathrm{Tr}(A|\psi\rangle\langle\psi|) = \sum_i \langle i| A |\psi\rangle\langle\psi| i\rangle = \langle\psi| A |\psi\rangle$$

$\hookrightarrow$ $\mathrm{Tr}(A) = \sum_i A_{ii}$

$\langle A, B \rangle = Tr(A^\dagger B) \rightarrow$ Hilbert-Schmidt / Trace Inner Product

$[A, B] = AB - BA \rightarrow$ Commutator

$\{A, B\} = AB + BA \rightarrow$ Anti-commutator

$\rightarrow$ Polar Decomposition

$A = UJ = KU \qquad \rightarrow U$ exists

$J \equiv \sqrt{A^\dagger A} \qquad K \equiv \sqrt{A A^\dagger}$

Basis of Quantum Information

Single Systems

Quantum States → represented by density matrices

Deterministic Operations

$$M|a\rangle = |f(a)\rangle \qquad a \in \Sigma$$

$$f : \Sigma \to \Sigma$$

a	$f(a)$
0	1
1	0

$$M = \sum_{a\in\Sigma} |f(a)\rangle\langle a|$$

$X|0\rangle = |1\rangle$

$X|1\rangle = |0\rangle$

$$X = |1\rangle\langle 0| + |0\rangle\langle 1| = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$M|a\rangle = |f(a)\rangle \rightarrow$

a	$f(a)$
0	3
1	0
2	2
3	1

$\rightarrow M = \boxed{\text{Ans.}}$

Dirac Notation

$\Sigma \in \{0 \_\_\_ n\}$

$|0\rangle = \begin{pmatrix} 1 \\ \vdots \\ 0 \end{pmatrix}$ $\quad |1\rangle = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}$ $\quad |j\rangle = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix}$ — $(j-1)$th row

$\langle 0| = (1 \_\_\_ 0)$

$\langle 1| = (0\,1 \_\_\_ 0)$

$\langle j| = (0 \_\_ \underbrace{1}_{(j-1)\text{th column}} \_\_ 0)$

$$M|a\rangle = \left(\sum_{b \in \{} |f(b)\rangle\langle b| \right) |a\rangle$$

when $b \neq a \rightarrow \langle b|a\rangle = 0$

when $b = a \rightarrow \langle b|a\rangle = 1$

$\rightarrow$ Dirac Notation

$$M|a\rangle = |f(a)\rangle$$

Probabilistic Operation

$$P = \begin{pmatrix} 1 & 1/2 \\ 0 & 1/2 \end{pmatrix}$$

$\downarrow$

Stochastic Matrix

$\downarrow$

Sum of all columns is 1

$\downarrow$

net probability

$P|0\rangle = |0\rangle$

$P|1\rangle = \frac{1}{2}|0\rangle + \frac{1}{2}|1\rangle$

Qubit Unitary Operation

→ Pauli Operations

$I =$

$X = \sigma_x = \sigma_1 =$

$Y = \sigma_y = \sigma_2 =$

$Z = \sigma_2 = \sigma_3 =$

$X|0\rangle = |1\rangle$
$X|1\rangle = |0\rangle$

↪ Bit flip

$Z|0\rangle = |0\rangle$
$Z|1\rangle = -|1\rangle$

phase flip

$XX = YY = ZZ = I$

$XY = iZ$
$YZ = iX$
$ZX = iY$

$\sigma_i \cdot \sigma_j = -\sigma_j \sigma_i \qquad i \neq j \qquad i,j \in \{1,2,3\}$

Hadamard Operation

$H =$

$H|0\rangle = \quad H|+\rangle = $

$H|1\rangle = \quad H|-\rangle = $

Phase Operations

$$P_\theta = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{pmatrix}$$

$S = P_{\pi/2} \rightarrow e^{i\pi/2} = i$

$T = P_{\pi/4} \rightarrow e^{i\pi/4} = \frac{1+i}{\sqrt{2}}$

$X = (HSH)^2$

$P_\theta|0\rangle = |0\rangle \qquad P_\theta|1\rangle = e^{i\theta}|1\rangle$

Multiple Qubit Systems

$$|0\rangle \otimes |0\rangle = |00\rangle$$

$$|0\rangle \otimes |1\rangle = |01\rangle$$

$$|1\rangle \otimes |0\rangle = |10\rangle$$

$$|1\rangle \otimes |1\rangle = |11\rangle$$

Probabilistic States

$$|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$$

$$\begin{pmatrix} \alpha^2 \\ \beta^2 \\ \delta^2 \\ \gamma^2 \end{pmatrix} \rightarrow \text{Stochastic Matrix}$$

Tensor Product

$$\rightarrow \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \otimes \begin{pmatrix} \delta \\ \gamma \end{pmatrix} = \begin{pmatrix} \alpha\delta \\ \alpha\gamma \\ \beta\delta \\ \beta\gamma \end{pmatrix}$$

$$(A \otimes C)(B \otimes D) = AB \otimes CD$$

$$\begin{pmatrix} a_{00} & a_{01} \\ a_{10} & a_{11} \end{pmatrix} \otimes \begin{pmatrix} b_{00} & b_{01} \\ b_{10} & b_{11} \end{pmatrix}$$

$$= \begin{pmatrix} a_{00}b_{00} & a_{00}b_{01} & a_{01}b_{00} & a_{01}b_{01} \\ a_{00}b_{10} & a_{00}b_{11} & a_{01}b_{10} & a_{01}b_{11} \\ a_{10}b_{00} & a_{10}b_{01} & a_{11}b_{00} & a_{11}b_{01} \\ a_{10}b_{10} & a_{10}b_{11} & a_{11}b_{10} & a_{11}b_{11} \end{pmatrix}$$

$$\langle a_1 \_\_ a_n | M_1 \otimes \_\_\_ \otimes M_n | b_1 \_\_ b_n \rangle$$

$$= \langle a_1 | M_1 | b_1 \rangle \_\_\_\_\_\_ \langle a_n | M_n | b_n \rangle$$

Quantum States

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

$|\phi\rangle = \gamma|0\rangle + \delta|1\rangle$

Product State $\rightarrow$ $|\psi\rangle \otimes |\phi\rangle$

All states aren't product states $\rightarrow$ Ex. & Bell Basis

Bell Basis $\rightarrow$ $|\phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$

$|\phi^-\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}}$

$|\psi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}$

$|\psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}$

GHZ State $\rightarrow$ $\frac{|000\rangle + |111\rangle}{\sqrt{2}}$

W State $\rightarrow$ $\frac{|001\rangle + |020\rangle + |100\rangle}{\sqrt{3}}$

$$|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle \rightarrow |a, b\rangle$$

$$P(a=0) = \sqrt{|\alpha|^2 + |\beta|^2}$$

Posterior State $\rightarrow \dfrac{\alpha|0\rangle + \beta|1\rangle}{\sqrt{|\alpha|^2 + |\beta|^2}}$

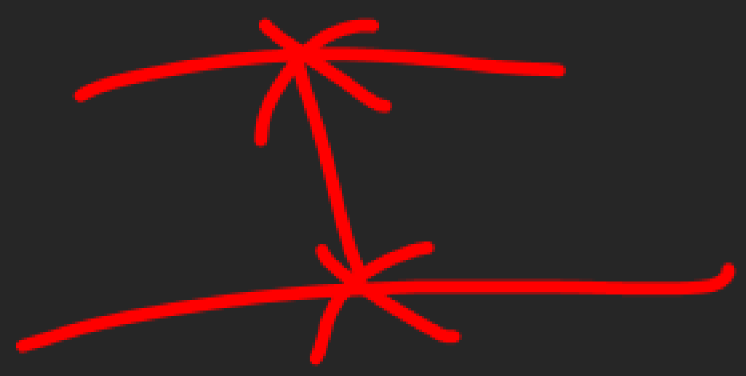
Unitary Operations on Multiple Qubits

$$(H \otimes I)|00\rangle = H|0\rangle \otimes |0\rangle = |+\rangle \otimes |0\rangle = \frac{|00\rangle + |10\rangle}{2}$$

SWAP Gate

$$SWAP|\psi \otimes \phi\rangle = |\phi \otimes \psi\rangle$$

$$\text{SWAP} = \sum_{a,b\in\Sigma} |a\rangle\langle b| \otimes |b\rangle\langle a|$$

$$\text{SWAP} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \rightarrow$$ 

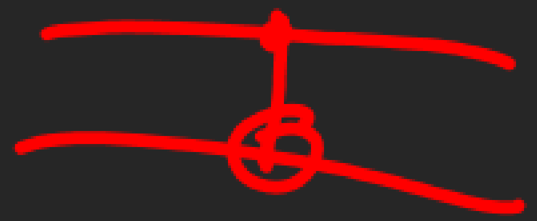
Controlled U Operation

On (X, Y) $\rightarrow |0\rangle\langle 0| \otimes I_Y + |1\rangle\langle 1| \otimes U = \begin{pmatrix} I_Y & 0 \\ 0 & U \end{pmatrix}$

↳ If 1st qubit is controlled-qubit

$$On(x,y) \rightarrow I_y \otimes |0\rangle\langle 0| + U \otimes |1\rangle\langle 1|$$

$$CNOT \rightarrow \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$



$$CNOT \rightarrow \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

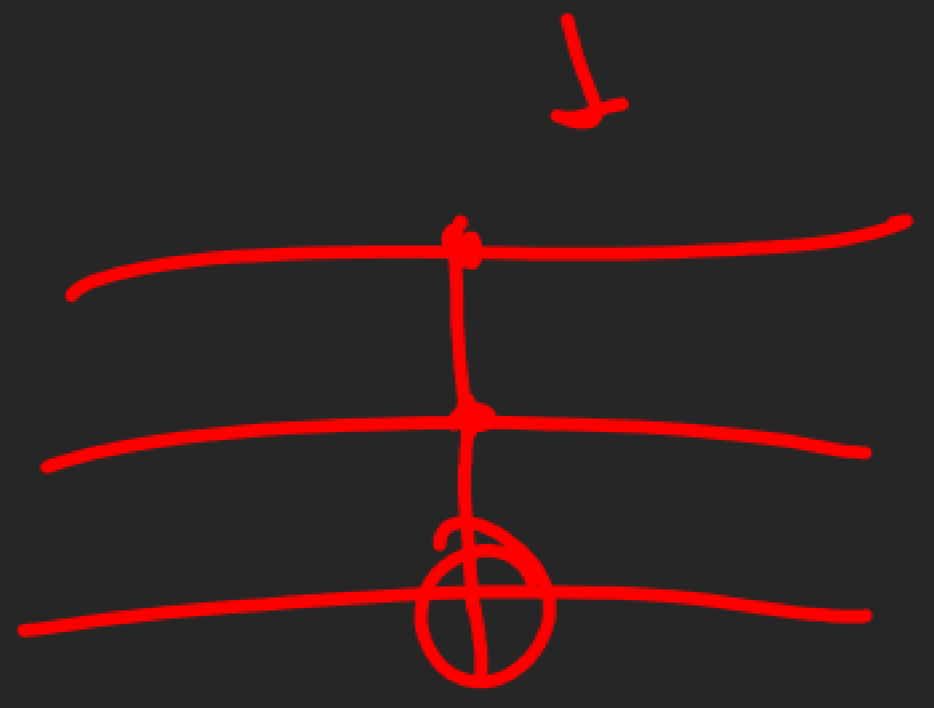
C-SWAP / Fredkin Gate Operation

$$|0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes SWAP$$

$$= \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

C-CNOT / Toffoli Gate / Operation

$$|0\rangle\langle 0| \otimes I_4 + |1\rangle\langle 1| \otimes CNOT$$



$$= \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$